Exercise 2.1.2 Suppose $\gamma = \{ x^a(u) \}$, a curve parameterized by u, is an affinely parameterized geodesic:

$$\ddot{x}^a + \Gamma^a_{bc}\, \dot{x}^b \dot{x}^c = 0 \qquad (2.12)$$

where

$$\dot{x}^a \equiv \frac{dx^a}{du}. \qquad (1)$$

Since $g_{ab} = g_{ab}(x^c) = g_{ab}(u)$ and $g^{ab} = g^{ab}(u)$, then also

$$\Gamma^a_{bc} = \frac{1}{2} g^{ad}(\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc}) = \Gamma^a_{bc}(u) \qquad (2.13)$$

Denote $\partial_a \equiv \dfrac{\partial}{\partial x^a}$.

Let L = Length of the tangent vector $\dot{x}^a$; i.e., $L(u)$ = Length of $\dot{x}^a(u)$.

(a) Show $\pm L^2 = g_{ab}\dot{x}^a\dot{x}^b$

(b) Express $\ddot{x}^a$ in expression (2.12) in terms of Γ^a_{bc} and $\dot{x}^a$

(c) Show L is constant

Solution:

(a) $L \overset{(1.80)}{=} \sqrt{|\, g_{ab}\, \dot{x}^a\, \dot{x}^b \,|} \quad \Leftrightarrow \quad L^2 = |\, g_{ab}\, \dot{x}^a\, \dot{x}^b \,|$

$\Leftrightarrow \quad \pm L^2 = g_{ab}\dot{x}^a\dot{x}^b \qquad ✓$

(b) Differentiating both sides:

LHS: $\dfrac{dL^2}{du} = \pm\, 2L\dot{L}$

RHS: $\dfrac{d}{du}(g_{ab}\dot{x}^a\dot{x}^b) = \dot{g}_{ab}\dot{x}^a\dot{x}^b + g_{ab}\,(\dot{x}^a\ddot{x}^b + \ddot{x}^a\dot{x}^b)$

$= \dot{g}_{ab}\dot{x}^a\dot{x}^b + 2g_{ab}\,\dot{x}^b\ddot{x}^a$

$$\Rightarrow \quad \pm\, 2L\dot{L} = \dot{g}_{ab}\dot{x}^a\dot{x}^b + 2g_{ab}\,\dot{x}^b\ddot{x}^a \qquad (2)$$

$$\dot{g}_{ab} \overset{(1)}{=} \frac{dg_{ab}}{du} = \frac{\partial g_{ab}}{\partial x^c}\frac{dx^c}{du} \overset{(1)}{=} (\partial_c g_{ab})\, \dot{x}^c \qquad (3)$$

$$\ddot{x}^a \overset{(2.12)}{=} -\, \Gamma^a_{cd}\, \dot{x}^c\dot{x}^d \quad ✓ \qquad (4)$$

(c) $2g_{ab}\,\dot{x}^b\ddot{x}^a \overset{(4)}{=} -2g_{ab}\dot{x}^b\,\Gamma^a_{cd}\,\dot{x}^c\dot{x}^d$

$$\Rightarrow\; 2g_{ab}\ddot{x}^a\dot{x}^b = -\,2g_{ab}\Gamma^a_{cd}\,\dot{x}^c\dot{x}^d\dot{x}^b \tag{5}$$

$$\pm\,2L\dot{L} \overset{(2)}{=} \dot{g}_{ab}\dot{x}^a\dot{x}^b + 2g_{ab}\,\dot{x}^b\ddot{x}^a \overset{(4)}{=} \dot{g}_{ab}\dot{x}^a\dot{x}^b - 2g_{ab}\Gamma^a_{cd}\,\dot{x}^c\dot{x}^d\dot{x}^b$$

$$\overset{(3)}{=} (\partial_c g_{\mathrm{ab}})\,\dot{x}^c\dot{x}^a\dot{x}^b - 2g_{ab}\Gamma^a_{cd}\,\dot{x}^c\dot{x}^d\dot{x}^b \tag{6}$$

$$\Gamma^a_{b\,c} \overset{(2.13)}{=} \frac{1}{2}g^{ad}(\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc})$$

$$\mathrm{b}\to\mathrm{c}\to\mathrm{d}\to\mathrm{e} \;\Rightarrow\; \Gamma^a_{c\,d} \overset{(2.13)}{=} \frac{1}{2}g^{ae}(\partial_c g_{ed} + \partial_d g_{ce} - \partial_e g_{cd}) \tag{7}$$

$$\therefore\; \pm\,2L\dot{L} \overset{(6)}{=} (\partial_c g_{\mathrm{ab}})\,\dot{x}^c\dot{x}^a\dot{x}^b - 2g_{ab}\Gamma^a_{cd}\,\dot{x}^c\dot{x}^d\dot{x}^b$$

$$\overset{(7)}{=} (\partial_c g_{\mathrm{ab}})\,\dot{x}^c\dot{x}^a\dot{x}^b - g_{ab}g^{ae}(\partial_c g_{ed} + \partial_d g_{ce} - \partial_e g_{cd})\,\dot{x}^c\dot{x}^d\dot{x}^b$$

$$= (\partial_c g_{\mathrm{ab}})\,\dot{x}^c\dot{x}^a\dot{x}^b - \delta^e_b\,(\partial_c g_{ed} + \partial_d g_{ce} - \partial_e g_{cd})\,\dot{x}^c\dot{x}^d\dot{x}^b$$

$$= (\partial_c g_{\mathrm{ab}})\,\dot{x}^c\dot{x}^a\dot{x}^b - (\partial_c g_{bd} + \partial_d g_{cb} - \partial_b g_{cd})\,\dot{x}^c\dot{x}^d\dot{x}^b$$

$$\overset{d\to a}{=} [\,(\partial_c g_{\mathrm{ab}} - \partial_c g_{ba}) + \partial_b g_{ca}\,]\,\dot{x}^c\dot{x}^a\dot{x}^b - \partial_a g_{cb}\dot{x}^c\dot{x}^a\dot{x}^b$$

$$\overset{a\leftrightarrow b}{=} \partial_b g_{ca}\,\dot{x}^c\dot{x}^a\dot{x}^b - \partial_b g_{ca}\,\dot{x}^c\dot{x}^b\dot{x}^a$$

$$= 0$$

$$\Rightarrow\quad \frac{\mathrm{d}L}{\mathrm{d}u} \overset{(a)}{=} \dot{L} = 0 \;\Rightarrow\; L \text{ is a constant} \quad ■$$